Beyond Reconstruction: Functional Neutralization of Channel Attention and Conditional Predictive Utility in ESPI Denoising

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Abstract

Electronic Speckle Pattern Interferometry (ESPI) produces spatial interference patterns used to characterize vibration modes; restoration of these images can alter the evidence available to a downstream classifier. We examine two connected questions: when Efficient Channel Attention (ECA) becomes functionally negligible inside normalized residual denoisers, and whether a denoised image adds predictive utility beyond excitation frequency. The study consolidates an earlier ECA/NoECA diagnostic, a corrected five-seed stress evaluation, a matched residual U-Net attention–normalization experiment, output-contract failure tests, and a nested frequency-conditional endpoint. The latter uses 11,706 real physical repeats across six specimens and five classes. A matched frequency-only probe achieves board-balanced Macro-F1 0.8167; adding raw images yields 0.8504, whereas adding V4R- or V5R-denoised images yields 0.8006 and 0.7747. Raw exceeds V4R on all six held-out boards in a paired descriptive comparison (mean difference 0.0497) and exceeds V5R on five (0.0756), even though the pre-specified absolute incremental-value decision remains inconclusive. A deterministic nearest-frequency diagnostic reaches 0.8612 and further exposes the strength of frequency information. We develop a mathematical analysis of scale normalization, grouped-channel effects, convolutional reparameterization, gate variability, and residual-path attenuation, specifying the conditions under which attention is exactly or approximately neutralized. The results identify an evaluation requirement: restoration quality, in-distribution classification, and conditional held-out-specimen utility are distinct quantities and should be measured separately. The remaining physical identity of the conditional image signal is an explicit measurement question.

Keywords: Electronic Speckle Pattern Interferometry; image restoration; Efficient Channel Attention; DnCNN; normalization; functional equivalence; conditional predictive utility; grouped evaluation; scientific imaging.

1 Introduction

Images acquired for physical measurement carry structured information whose meaning depends on the acquisition process. In Electronic Speckle Pattern Interferometry (ESPI), coherent illumination and interference yield a spatial pattern modulated by deformation and experimental geometry. Interference fringes and their spatial organization can help distinguish vibration states. Speckle, illumination fluctuations, limited contrast, and repeat-to-repeat variation complicate direct analysis. This setting makes image restoration consequential: a denoiser changes the evidence on which the subsequent inference procedure operates.

The conventional restoration objective evaluates correspondence to a target image. In experiments where the target is an averaged acquisition, that target is itself an empirical operator applied to repeated observations. Averaging can suppress stochastic variation, while variation in phase, registration, or interference contrast can also alter the structures of interest. A model that fits the averaged proxy therefore needs an independent test of downstream usefulness. This concern extends beyond ESPI to restoration in scientific imaging whenever a target is constructed from imperfect repeated measurements [9, 7].

Attention introduces a second evaluation problem. Efficient Channel Attention (ECA) [2] assigns data-dependent multiplicative weights to feature channels. A nonconstant gate is an observable internal computation, yet that computation may have little effect on the final denoised image. Normalization and subsequent convolutions provide mechanisms through which channel scaling can be canceled, approximated, or redistributed. Moreover, a residual denoiser can propagate its input through a direct identity path, leaving the final mapping nearly unchanged even when intermediate gates vary.

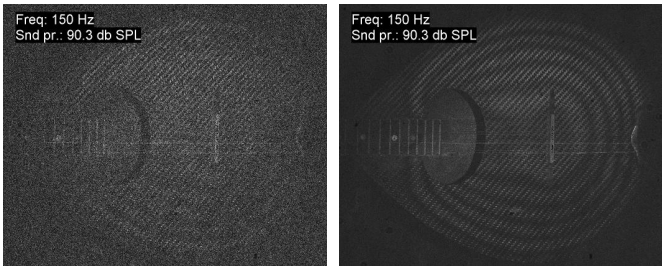
This extended manuscript organizes the completed evidence around three research questions.

RQ1. Functional neutralization. Under what algebraic and statistical conditions can active channel attention have an undetectable end-to-end effect?

RQ2. Architectural and supervision effects. How do supervision, normalization, and evaluation regime alter the observed utility of DnCNN-ECA and restoration backbones?

RQ3. Conditional image utility. Does a particular image view improve classification *conditional on excitation frequency* when tested on held-out physical specimens?

An earlier short-paper investigation of DnCNN-Lite, pseudo-synthetic supervision, and real-aligned supervision motivated a broader audit. Its preliminary robustness interpretation rested on a three-run launcher with incomplete seed propagation. The ensuing revision corrected the seed protocol, compared



(a) Single-shot measurement **(b)** Supplied averaged proxy

Figure 1: Two examples taken from the original experimental manuscript (carbon specimen C01, excitation near 150 Hz). The acquisition and averaged-proxy views illustrate the distinction between measured texture and a constructed reference; visual smoothness alone does not establish preservation of vibration-mode information.

residual U-Net variants under BatchNorm and GroupNorm, and inspected gate interventions. More recent experiments revealed an additional confound: *excitation frequency itself strongly predicts the five-class endpoint*. High image-only performance cannot establish added image value once that predictor is available.

The contributions are a unified mathematical account with explicit applicability conditions; a separation of three evaluation estimands; a corrected and controlled evidence synthesis; and a nested frequency-conditional analysis that changes the interpretation of the earlier denoising comparisons. This is a methods-and-evaluation study. Proposed additional restoration objectives are identified as future experimental hypotheses, with no completed-model performance attributed to them.

2 ESPI measurements and the prediction problem

2.1 From optical measurements to image evidence

An idealized interference intensity at pixel p can be written as

$$I(p) = I_{\mathrm{dc}}(p) + A(p)\cos\big(\phi_{\mathrm{ref}}(p) + \Delta\phi(p)\big) + \eta(p), \quad (1)$$

where I_{dc} is background intensity, A is the modulation amplitude, ϕ_{ref} is the optical phase offset, $\Delta\phi$ contains the displacement-sensitive phase change, and η aggregates unresolved acquisition effects. Equation (1) is an illustrative local interferometric model: the real dataset contains processed fringe images and does not supply an independently calibrated $\Delta\phi$ field for every observation. Time-averaged configurations can involve a different nonlinear response, including Bessel-type modulation; therefore grayscale contrast cannot generally be equated with displacement amplitude.

Fringe spacing, curvature, topology, regional contrast, and the location of nodal-like structures can be useful to a learned classifier. Their relation to the underlying physical state depends on illumination geometry, phase processing, and acquisition conditions. This paper evaluates *classification utility* and does not infer calibrated physical displacement from classifier performance.

2.2 Observation, condition, view, and label

Let B denote physical specimen, C an excitation/acquisition condition, F excitation frequency, R repeat identity, X the measured single-shot image, and Y the five-class annotation. Let D_j denote a specified pretrained denoiser, with image view

$$V_j = D_j(X), \qquad V_0 = X \quad \text{(Raw)}. \quad (2)$$

The two evaluated denoising views are V_{4R} and V_{5R}. Each image view is paired sample-wise with the same F and Y in the primary conditional experiment.

The acquisition inventory documented in the earlier revision comprises six boards (carbon C01–C03, wood W01–W03), 23,769 single-shot PNG acquisitions, 1,413 supplied averaged proxies, and a mixed real-aligned/pseudo pair cache. These counts describe distinct historical data products and should not be substituted for the primary frozen endpoint, which contains 11,706 *unique real physical repeats* with complete five-class support. The original five-class classifier inventory, also used in earlier experiments, comprised 13,278 labeled images; counts therefore differ by evaluation contract.

2.3 Why the proxy and label require provenance

For repeated measurements $X_1, \ldots, X_m$ of a condition, an averaged proxy may be represented abstractly as

$$\widetilde{X}_C = \mathcal{A}(X_1, \ldots, X_m; \zeta), \quad (3)$$

where $\mathcal{A}$ includes unknown historical image combination and ζ includes possible processing settings. The public audit establishes incomplete provenance of this historical operator. Consequently, $\widetilde{X}_C$ is an *operational supervision reference*, while its status as a unique clean optical measurement remains unverified.

The class labels are heavily predictable from F. This is not automatically a leakage defect, but it changes the scientific estimand: image-only mode classification can primarily reflect a frequency–class association. Accordingly, the conditional experiment explicitly measures improvement over a matched frequency-only model. The independent physical or annotational basis of the five classes remains important for interpreting any conditional gain.

3 Prior work and scientific positioning

DnCNN introduced residual-learning formulations for image denoising [1]. ECA [2] and related channel-attention mechanisms such as SE and CBAM [5, 6] motivate low-cost modulation of intermediate features. Batch Normalization (BN) [3] and Group Normalization (GN) [4] alter the scale geometry of the learned representation. Interferometric denoising has been studied using CNN restoration [7] and attention-augmented DnCNN approaches [8]; methods for denoising without conventional clean targets include Noise2Noise and Noise2Void [9, 10]. More recent restoration backbones, including NAFNet [11], provide a relevant architectural stress test.

Our focus is the *evaluation relation* between restoration and downstream information, alongside an explicit account of when

architectural attention can become functionally inconsequential. We distinguish three questions that have sometimes been blended: whether a denoiser matches an averaged reference, whether its output supports image-only recognition, and whether its output adds information beyond a strong non-image baseline on unseen specimens. These are different tasks with different validation units.

4 Architectures, supervision, and evaluation contracts

4.1 Residual DnCNN and attention placement

Write the denoiser as

$$\widehat{X} = D_\theta(X) = X - R_\theta(X), \tag{4}$$

where R_θ predicts a residual. DnCNN-Lite uses a 17-layer convolutional stack with 32 feature channels. The corrected real-aligned V4R model uses GroupNorm(8) and ECA insertion at positions $[0, 1, 2]$. The V5R model uses GroupNorm(8) and positions $[0, 1, 2, 3, 6, 10, 14]$. Both are explicitly *attention-containing*: the V5R–V4R contrast probes placement density and configuration, not the isolated causal effect of adding ECA.

The earlier pseudo-supervised V3 fair A/B experiment compared ECA and no-ECA variants under locked settings, and included an observed near-float-level equivalence. The original proposed explanation was normalization-mediated cancellation and convolutional absorption; the mathematical section below sharpens this explanation and specifies what the observation establishes.

4.2 Supervision regimes

The pseudo regime uses synthetic or pseudo-noisy input/reference pairs whose nuisance statistics need not match real speckle acquisition. The real-aligned regime pairs a real single-shot with an averaged proxy from the same excitation condition. A reconstruction objective used in the corrected DnCNN program was a weighted Charbonnier and mean-squared-error combination,

$$\mathcal{L}_{\text{rec}} = 0.7\,\mathcal{L}_{\text{Charb}} + 0.3\,\mathcal{L}_{\text{MSE}}, \tag{5}$$

with the averaged proxy as reference. The historical data cache additionally contained 243 pseudo pairs; its exact composition is stated rather than implicitly equating all cached pairs to real acquisitions.

4.3 Three noninterchangeable estimands

The corrected five-seed stress experiment controls seeds $42, 13, 37, 101, 202$ across Python, NumPy, PyTorch CPU/CUDA, data splitting, augmentation, additive stress noise, and classifier training/checkpoint selection. In the earlier revision, the denoisers ran for 20 epochs with 256×256 crops, batch size 8, AdamW learning rate 2×10^{-4} and cosine annealing. The downstream ResNet-18 classifier used one grayscale input channel, five output classes, class-balanced sampling, AdamW, and validation Macro-F1 checkpoint selection. The severe Gaussian stress has $\sigma = 25$ on an 8-bit intensity scale and is a stress diagnostic rather than a generative model of nominal speckle.

The separate residual U-Net-Lite sensitivity uses widths $8, 16, 32$, a 64-channel bottleneck, ECA3 inserted after enc0/enc1/enc2 where applicable, and matched seed-42 controls. Each 2×2 combination of GN/BN and ECA/no-ECA was compared at the same epoch (15). This supports a carefully scoped architecture–normalization interaction analysis; the six held-out boards do not create independent training-seed replicates.

The nested endpoint is more demanding. For each board b, view j, and seed s, image logits for the five development boards are produced by *condition-grouped inner out-of-fold* (OOF) models. The frequency-only and frequency-plus-image fusion heads use the same balanced multinomial logistic regression family, trained on OOF records only. A single outer board is held out for evaluation. The completed inventory records 225 newly fitted inner models, 72 new outer models, 18 verified historical outer models reused, and 90 outer prediction/90 inner OOF tables. Audits found exact coverage and no recorded held-out-board/condition-group leakage. Published tables can be validated for counts and hashes; the private training tensors and checkpoints are not distributed.

5 Mathematical analysis of functional attention neutralization

5.1 ECA as a data-dependent diagonal operator

Let $H(X) \in \mathbb{R}^{C\times S}$ denote a flattened feature map with $S = HW$ spatial positions. Global average pooling P yields $m(H) \in \mathbb{R}^C$ with

$$m_c(H) = S^{-1} \sum_{p=1}^{S} H_{cp}. \tag{6}$$

With a banded one-dimensional channel convolution matrix $K \in \mathbb{R}^{C\times C}$, ECA produces

$$\begin{aligned} g(H) &= \sigma\big(Km(H)\big), \\ T(H) &= G(H)H, \quad G(H) = \operatorname{diag}(g(H)), \end{aligned} \tag{7}$$

where $0 < g_c < 1$ for the usual sigmoid gate. A centered-gain variant changes the gate parametrization and should be distinguished from conventional sigmoid ECA in implementation claims. A nontrivial variance of $g_c(H)$ establishes gate activity; the observable end-to-end effect depends on all following operations.

For an input X, define a no-ECA network $f_0(X)$ and an ECA network $f_1(X)$. Their observed functional discrepancy over a fixed evaluation set $\mathcal{T}$ is

$$\mathcal{D}_1(\mathcal{T}) = |\mathcal{T}|^{-1} \sum_{X\in\mathcal{T}} \|f_1(X) - f_0(X)\|_1 / N, \tag{8}$$

with N output pixels. The earlier V3 diagnostic found output differences on the order of floating-point precision in its audited configuration, despite active gates. This is an *empirical*

Table 1: Interpretation boundaries for the three completed evaluation families. The numerators, split structures, and scientific questions differ.

Evaluation family	Unit and split	Evaluated quantity	Main interpretation
Corrected five-seed stress	Seed-level repeated training; random in-distribution split	Image-only Macro-F1 after severe additive-noise corruption	Training/evaluation robustness within the stress protocol
Historical six-board transfer	One held-out board with other boards for classifier development; seed-42 reference	Image-only Macro-F1 by physical board	Physical-board heterogeneity for that pipeline; denoiser exposure is not independently established as held-out
Nested conditional endpoint	Held-out board, five image seeds, condition-grouped inner OOF fusion	$\mathrm{F1}(F, V_j) - \mathrm{F1}(F)$	Incremental predictive utility of each view beyond a matched frequency-only model

functional-equivalence observation over the evaluated inputs, distinct from a theorem asserting parameter identity or global equivalence over all possible images.

5.2 Batch Normalization: an exact invariance under restricted scaling

Consider a single feature channel over the reduction set $\mathcal{A}$ used for BN, with sample mean μ, population-style batch variance v, learned affine parameters γ, β, and numerical stabilizer $\epsilon > 0$:

$$\mathrm{BN}(h) = \gamma \frac{h - \mu}{\sqrt{v + \epsilon}} + \beta. \tag{9}$$

Assume that a *single positive scalar* $a > 0$ multiplies *every element* of this channel throughout $\mathcal{A}$; the statistics are recomputed from the scaled activations. Then $\mu(ah) = a\mu$ and $v(ah) = a^2 v$. Therefore

$$\mathrm{BN}(ah) = \gamma \frac{a(h - \mu)}{\sqrt{a^2 v + \epsilon}} + \beta \tag{10}$$

$$= \gamma \frac{h - \mu}{\sqrt{v + \epsilon/a^2}} + \beta. \tag{11}$$

For $\epsilon = 0$ and $v > 0$, this equals $\mathrm{BN}(h)$ exactly. With finite ϵ, their difference is

$$\Delta_{\mathrm{BN}}(h; a) = \gamma(h - \mu) \left[\frac{1}{\sqrt{v + \epsilon/a^2}} - \frac{1}{\sqrt{v + \epsilon}} \right]. \tag{12}$$

A first-order expansion at small ϵ/v gives

$$\Delta_{\mathrm{BN}} \approx \frac{\gamma(h - \mu)\epsilon}{2v^{3/2}} \left(1 - a^{-2}\right), \tag{13}$$

which makes the numerical rather than conceptual role of ϵ explicit.

Proposition 1 (BN scale invariance in training mode) *For positive channel scaling constant over the BN statistics-reduction set, and for nonzero channel variance, BN is invariant to that scaling when $\epsilon = 0$. For $\epsilon > 0$, the discrepancy is given by Eq.* (12)*.*

Important scope. ECA generates an *input-dependent* gate. In training-mode BN, distinct samples can have distinct gates; the scaling is then generally not constant over the minibatch reduction set, and the exact cancellation argument above does not apply. In evaluation mode, BN usually uses fixed running means and variances. Applying an ECA gate immediately before such a BN changes the output as

$$\mathrm{BN}_{\mathrm{eval}}(ah) - \mathrm{BN}_{\mathrm{eval}}(h) = \gamma \frac{(a - 1)h}{\sqrt{v_{\mathrm{run}} + \epsilon}}, \tag{14}$$

with no instantaneous batch-statistics cancellation. Consequently, a post hoc explanation of near-identical evaluated models must also consider learned reparameterization, small downstream sensitivity, and topology. The statement "BN automatically removes ECA" is mathematically incorrect without the stated conditions.

5.3 Group Normalization is only group-scale invariant

For one input and a group $\mathcal{G}$ of q channels, GN computes one mean and variance across the group's channel–spatial elements. Let $\overline{h}_{\mathcal{G}}$ and $v_{\mathcal{G}}$ denote these statistics. If *all* channels in the group are scaled by a shared positive factor $a_{\mathcal{G}}$, then the same algebra yields invariance at $\epsilon = 0$.

Per-channel ECA supplies a_c, typically different for $c \in \mathcal{G}$. In this case

$$\overline{Gh}_{\mathcal{G}} = \frac{1}{qS} \sum_{c \in \mathcal{G}} \sum_{p} a_c h_{cp}, \tag{15}$$

while

$$v_{\mathcal{G}}(Gh) = \frac{1}{qS} \sum_{c,p} \left(a_c h_{cp} - \overline{Gh}_{\mathcal{G}} \right)^2. \tag{16}$$

These generally cannot be factored into $a\overline{h}_{\mathcal{G}}$ and $a^2 v_{\mathcal{G}}$ for a common a. The normalized cross-channel contrasts change. GN also has per-channel learnable affine parameters, which permit learned compensation, but the presence of γ_c and β_c does not prove that compensation occurs.

Proposition 2 (Restricted GN invariance) *GN is invariant (apart from ϵ) to a positive multiplier shared by all elements of each normalization group. Independent positive channel multipliers inside the same group generally alter its normalized output.*

This distinction matters for GroupNorm(8) with 32 feature channels: each group contains four channels, whereas ECA gates are channel-specific. In the matched U-Net study, the

GN ECA arm had *higher* downstream Macro-F1 than GN no-ECA. GN is therefore compatible with useful, stable attention-dependent rescaling rather than universally erasing it.

5.4 Convolutional absorption: static gates and variable gates

Consider the linear operation immediately after attention, $Z(X) = WG(H(X))H(X)$, where W denotes the next convolution in matrix form. If $g(X)$ is constant across the relevant input set, $G(H(X)) = \overline{G}$, then

$$Z(X) = (W\overline{G})H(X) = W'H(X), \quad W' = W\overline{G}. \tag{17}$$

This is an exact reparameterization of a *static* diagonal channel scaling into the input-channel weights of a following linear convolution. It also applies to affine biases under the appropriate bookkeeping. The statement is existential: an equivalent parameterization *can* be constructed. It does not imply that separately trained ECA/no-ECA models necessarily learned the matching W'.

For input-dependent gates, write

$$G(X) = \overline{G} + \Delta G(X). \tag{18}$$

Using $W' = W\overline{G}$ leaves the residual

$$Z(X) - W'H(X) = W\Delta G(X)H(X). \tag{19}$$

Thus the gate's input-dependent component is the relevant term that cannot, in general, be absorbed by a fixed linear kernel. Its norm obeys

$$\|Z(X) - W'H(X)\|_2 \le \|W\|_2 \, \|\Delta G(X)\|_2 \, \|H(X)\|_2 . \tag{20}$$

If a downstream map Q is L_Q-Lipschitz on these activations, then

$$\begin{aligned} &\|Q(Z(X)) - Q(W'H(X))\|_2 \\ &\quad \le L_Q \, \|W\|_2 \, \|\Delta G(X)\|_2 \, \|H(X)\|_2 . \end{aligned} \tag{21}$$

The resulting neutralization is approximate when the variable gates are small, when the following convolution suppresses their directions, or when subsequent nonlinear computations are locally insensitive. The bound is diagnostic rather than a numerical explanation by itself: one would need $\|W\|$, gate variability, and local downstream sensitivity to make it quantitative.

5.5 The ECA Jacobian distinguishes gate activity from effective sensitivity

For clarity, vectorize H as $h \in \mathbb{R}^{CS}$ and define the pooling map $P : \mathbb{R}^{CS} \to \mathbb{R}^{C}$; $m = Ph$. Let $A(h)$ replicate $g = \sigma(KPh)$ across all spatial positions. Then the attention operation $t(h) = h \odot A(h)$ has Jacobian

$$J_t(h) = \operatorname{diag}(A(h)) + \operatorname{diag}(h)\, U \operatorname{diag}\big(\sigma'(KPh)\big) KP, \tag{22}$$

where U broadcasts a channel gate to its S spatial positions. The first term is direct channel rescaling; the second is the derivative of the *data dependence* of the gate. For the usual sigmoid, $\sigma'(u) \le 1/4$. Consequently, with a consistent operator norm,

$$\|J_t - \operatorname{diag}(A)\| \le \tfrac{1}{4} \, \|\operatorname{diag}(h)\| \, \|U\| \, \|K\| \, \|P\| . \tag{23}$$

A following W and downstream Jacobian J_Q give $J_{Q\circ W\circ t} = J_Q W J_t$. The variable-gate contribution can lie approximately in low-gain directions of $J_Q W$, even when the gates themselves change. Hence a nonzero gate variance is not equivalent to a large output sensitivity.

5.6 Residual pathways and downstream cancellation

For the denoiser in Eq. (4), two variants have difference

$$D_1(X) - D_0(X) = -\big(R_1(X) - R_0(X)\big). \tag{24}$$

The identity path can dominate the absolute output scale and make both reconstructions close to X. This closeness is not sufficient to establish that ECA was normalized away; it can also reflect small learned residuals, insensitive output kernels, or parameters that happen to produce similar residual predictions. Conversely, apparently modest pixel differences can be important if concentrated near rare discriminative structures. Effective-function audits therefore require pixel differences, relative residual differences, and downstream outputs under the same evaluation inputs.

5.7 Interventions: from correlation to mechanism

Four interventions distinguish possible explanations. At fixed learned weights, replace (i) learned gates by identity gates, (ii) learned gates by their per-channel calibration means, (iii) gates by sample-shuffled gates, and (iv) learned BN/GN affine coefficients by controlled alternatives. A constant-gate intervention tests how much dynamic modulation contributes beyond static scaling. Identity gates test total attention contribution for the *trained* network. These interventions differ from retraining an independent no-ECA model; they answer whether the fitted solution relies on the gate at inference time.

The completed residual U-Net audit reported that under GN the ECA benefit was largely compatible with stable scaling, while under BN dynamic gate perturbations could remain observable without producing a consistent task gain. The affine-parameter audit did not establish a simple inverse scaling law, so assigning the entire mechanism to γ and β would go beyond the evidence. Exact experiment-specific gate intervention values are retained in the prior audit package; the present manuscript does not manufacture unreported distributions or response curves.

6 Statistical design for conditional image utility

6.1 Conditional estimand and competing baselines

Let $\mathcal{A}_F$ be a fixed learner trained on frequency, and $\mathcal{A}_{F,V}$ the same-class fusion learner trained on frequency plus out-of-fold

image logits. Denote the evaluated board-balanced Macro-F1 by $M(\cdot)$; the image's operational incremental value is

$$\Delta_j = M\big(\mathcal{A}_{F,V_j}\big) - M\big(\mathcal{A}_F\big). \tag{25}$$

This is a *learner-, data-, and protocol-specific difference in predictive utility*, not an estimate of conditional mutual information $I(Y; V_j \mid F)$.

For held-out board b and image seed s, write

$$\Delta_{j,b,s} = M_{b,s}(F, V_j) - M_{b,s}(F), \tag{26}$$

with board-first mean

$$\widehat{\Delta}_j = \frac{1}{B}\sum_{b=1}^{B}\left[\frac{1}{S}\sum_{s=1}^{S}\Delta_{j,b,s}\right], \quad B = 6, S = 5. \tag{27}$$

A paired contrast between image views j and k cancels the common matched frequency comparator:

$$\widehat{\tau}_{j-k} = \frac{1}{B}\sum_{b}\frac{1}{S}\sum_{s}\big(\Delta_{j,b,s} - \Delta_{k,b,s}\big). \tag{28}$$

The held-out board is the appropriate unit for physical-domain sensitivity. Treating 6×5 board–seed cells as 30 independent boards would overstate independent physical replication.

6.2 Selection control and nested fusion

The fusion heads are fit only to inner OOF records from the development boards; outer held-out-board labels are reserved for evaluation. This protects the conditional difference against the most direct form of in-sample image-logit leakage. All image views share the same frozen identity map between real repeat, label, and frequency. The public audits cover duplicate and missing predictions, held-out-board exclusions, and condition-grouped inner model construction. Historical denoiser pretraining and the nested classifier/fusion procedure should be distinguished when describing full end-to-end physical generalization.

6.3 Decision rules and descriptive uncertainty

A frozen absolute decision required *both* mean $\widehat{\Delta}_j \geq 0.030$ and strictly positive board-level increments on at least five of six boards. In the completed endpoint, Raw passed the mean criterion but was positive on only four boards. The consequent frozen decision was `DNCNN_INCREMENTAL_VALUE_INCONCLUSIVE`; the later exploratory paired audit could not replace it.

The paired comparisons were conducted after this absolute decision using existing prediction tables and no new fitting. Board bootstrap percentile intervals and exact enumeration of $2^6 = 64$ sign assignments are useful descriptive summaries. Their conventional population interpretations require assumptions beyond the six purposively observed physical specimens. In particular, a low exact sign-flip value is not automatically a randomized experimental p-value without an appropriate exchangeability or randomization basis.

Table 2: Corrected five-seed in-distribution stress, mean ± sample SD. The units are fractions, not percentages.

Input view	Accuracy	Macro-F1
Raw	0.9328 ± 0.0158	0.8439 ± 0.0164
V4R	0.9254 ± 0.0178	0.8269 ± 0.0227
V5R	**0.9429 ± 0.0144**	**0.8589 ± 0.0314**

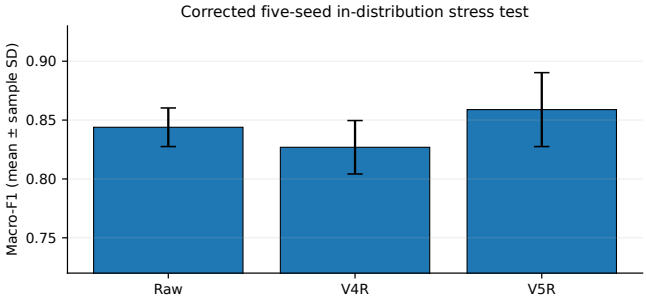


Figure 2: The corrected, seed-controlled severe-noise stress comparison. Error bars denote sample SD over five seeds. This diagnostic is not pooled with held-out-board endpoints.

7 Results

7.1 Corrected in-distribution stress: a revised ranking

The corrected five-seed replication reverses the old three-run pilot ranking: V5R exceeds V4R in all five paired runs. The mean V5R–V4R Macro-F1 difference is +0.0319 (reported small-n t interval $[0.0085, 0.0554]$), but exact two-sided sign-flip enumeration over five paired seeds has minimum nonzero granularity $2/32 = 0.0625$. V5R minus Raw is only +0.0150 on average, with an interval spanning zero. These results support relative V5R stress performance under this protocol without licensing a universal denoising benefit.

7.2 Historical image-only board transfer

The historical six-board classifier-transfer audit reports board-balanced Macro-F1 0.4970 for Raw, 0.4395 for V4R, and 0.4681 for V5R; corresponding mean Accuracies are 0.8314, 0.8294, and 0.8321. Board/material heterogeneity is substantial. This evaluates image-only representations under a grouped classifier protocol, and should be interpreted with the stated denoiser-pretraining exposure boundary. More importantly, frequency-conditional results below demonstrate why image-only transfer alone is insufficient for the five-class task.

7.3 ECA–normalization interaction in residual U-Net

The GN ECA–no-ECA difference is +0.1429 Macro-F1 and is positive on all six boards. The corresponding BN difference is −0.0144, positive on two of six. The paired interaction (GN effect minus BN effect) is +0.1573, with five of six board-level interactions positive; its exact six-board sign-flip diagnostic

Table 3: Exploratory matched residual U-Net-Lite results at fixed epoch 15 and seed 42 (board-balanced).

Arm	Accuracy	Macro-F1
GN, no ECA	0.5940	0.1833
GN, ECA	0.7525	0.3262
BN, no ECA	0.5348	0.2650
BN, ECA	0.6122	0.2506

Table 4: Primary conditional endpoint. Same held-out boards and matched frequency-only comparator; equal board weights. The nearest-frequency result is a separate diagnostic.

Learner / view	Macro-F1	Δ vs F-only
F-only matched	0.816656	–
Raw F+I	**0.850356**	+0.033699
V4R F+I	0.800619	−0.016037
V5R F+I	0.774716	−0.041940
Nearest-frequency	**0.861237**	separate

was reported as 0.0625. This does not validate a universal GN superiority theorem or a single normalization-cancellation mechanism. It demonstrates that the observed utility of channel attention is architecture- and normalization-dependent.

7.4 Output-contract failure in restoration backbones

The output-preservation audit quantified deviations from raw inputs. V4R and V5R had input–output correlations 0.7533 and 0.8400, and gradient-retention scores 0.166 and 0.323. The tested GN residual U-Net variants had negative input–output correlations, and the tested native-SCA NAFNet-Tiny had correlation 0.0634 with gradient retention 0.033. The NAFNet control produced near-white outputs and on the C01 downstream probe had Accuracy 0.0901 and Macro-F1 0.0778. These are observations about *specific trained configurations* and a particular averaged-proxy contract; they establish a restoration-to-recognition failure mode in this experiment, not an architecture-wide ordering.

7.5 Nested frequency-conditional endpoint

The principal finding is a ranking change after conditioning on frequency. Raw F+I exceeds the matched F-only logistic baseline by 0.033699 Macro-F1. The V4R and V5R views decrease this score by 0.016037 and 0.041940, respectively. Raw meets the frozen average-effect threshold 0.030 but improves only four of the six held-out boards, so the absolute criterion is inconclusive. Both denoised views fail its effect and consistency gates. The deterministic nearest-frequency diagnostic reaches 0.861237 Macro-F1; this limits any claim that the raw-image fusion learner outperforms frequency-only prediction in general.

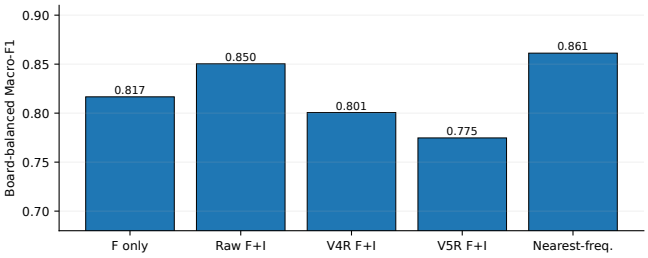


Figure 3: Board-balanced Macro-F1 in the frozen five-class endpoint. Raw F+I beats the *matched* logistic F-only baseline, whereas nearest-frequency (a distinct deterministic diagnostic) is stronger than Raw F+I. No universal advantage over all frequency-only alternatives follows.

Table 5: Frozen descriptive board-first contrasts: Raw minus denoised conditional increments. The intervals are descriptive board bootstrap summaries, not population-level confidence guarantees.

Contrast	Mean	+ boards	95% interval	Sign flip
Raw–V4R	+.049737	6/6	[.03141,.06840]	.03125
Raw–V5R	+.075639	5/6	[.03514,.10917]	.06250
V4R–V5R	+.025903	4/6	[-.01782,.06351]	.31250

7.6 Board-first paired comparisons

The paired audit found Raw greater than V4R on six of six boards and greater than V5R on five of six; the condition-balanced sensitivity preserved both directional counts. The strongest class-level differences are concentrated in `1_2` and `2_1`, as shown in Fig. 5. This localization is a *class-prediction effect*, not identification of a particular fringe family or calibrated vibration feature. The claim justified by these analyses is that Raw provides greater conditional predictive utility under the locked learner and endpoint.

7.7 Why the three rankings disagree

The earlier stress experiment perturbs raw test images with strong Gaussian noise in an in-distribution setting, for which V5R shows a relative advantage over V4R. Historical board-transfer tests evaluate image-only classification on distinct physical specimens and favor Raw in mean Macro-F1. The nested endpoint asks the more restrictive question of what image logits contribute after F is already available, and again favors Raw over the denoised views. These outcomes address different conditional distributions and model-comparison questions, and their divergence is expected rather than contradictory.

8 Discussion: what is measured, and what remains physical

8.1 Functional equivalence is a system property

The V3 float-level comparison motivates mechanistic analysis, but its observed agreement is not a universal theorem about BN, GN, or ECA. The exact BN statement requires a gate constant across a statistics-reduction set. GN is only invariant to scaling common to the channels in each group. Absorption

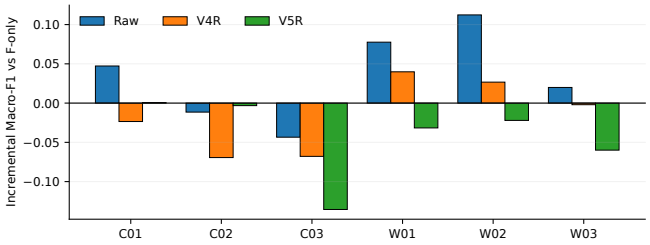


Figure 4: Six held-out-board increments over the matched frequency-only comparator; seeds are averaged within board. The physical specimen is the displayed unit.

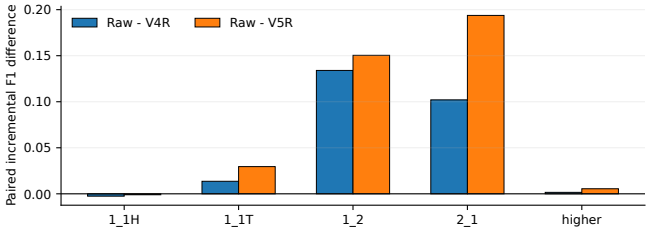


Figure 5: Board-balanced class-level paired differences in *incremental* F1. The largest Raw advantages relative to denoised views occur in classes `1_2` and `2_1`; support is unequal across classes and boards.

by a following convolution is exact for constant gates and approximate for data-dependent gates. Residual-path dominance and low downstream Jacobian sensitivity provide additional pathways to small end-to-end effects. Together these results explain how active gates and nearly identical outputs can coexist while preserving the need for intervention-based mechanism identification.

8.2 The relevant control is a strong non-image predictor

The frequency-only Macro-F1 0.8167 and nearest-frequency score 0.8612 materially change the interpretation of high image-only accuracies. The analysis now evaluates image contribution *conditional on a matched frequency learner*, and it reports the strongest simpler diagnostic rather than suppressing it. In the physical setting, excitation frequency is directly linked to experimental state selection. A model can learn class decisions from experimental conditions without necessarily decoding complex fringe structure. Thus, a convincing claim that denoising preserves scientifically interpretable image information calls for additional independent observables, counterfactual frequency/condition tests, or controlled labeling evidence.

8.3 What the observed Raw advantage establishes

The completed paired audit demonstrates a robust directional descriptive contrast across the six available boards. It does not determine whether the useful residual signal is modal fringe geometry, acquisition artifacts correlated with labels, or another structured image feature. The relevant empirical question is identifiable using new acquisition/provenance information: if an independently measured vibration-state variable is supplied, one can evaluate whether predicted physical quantities improve when trustworthy image descriptors are added to experimental metadata. That is a future experiment and is not presented here as a completed result.

8.4 Why averaged-proxy restoration can fail

Consider a stylized fringe signal with a shared amplitude a but repeat-dependent phase error δ_r,

$$X_r(p) = b(p) + a(p)\cos\big(\phi(p) + \delta_r(p)\big) + \eta_r(p). \qquad (29)$$

If δ_r are identically distributed, the amplitude of the averaged fringe term is multiplied by $|\mathbb{E}[e^{i\delta_r}]|$. For centered Gaussian phase jitter of variance σ_δ^2, this factor is $e^{-\sigma_\delta^2/2}$. Thus a smoother average may carry a weaker carrier/fringe modulation even when random additive noise decreases. This is a mathematical counterexample to the assumption that a mean image is always a signal-preserving reference. It does not establish that this precise phase-jitter process generated the archived ESPI proxies; their historical construction is incompletely documented.

With squared error, the optimal population regressor predicts $D^*(x) = \mathbb{E}[\widetilde{X}_C \mid X = x]$, not the unobserved calibrated displacement field. If the target $\widetilde{X}_C$ systematically suppresses some mode-discriminative variation, low reconstruction risk can coexist with low downstream utility. This provides a coherent interpretation of the observed whitening failure under its specific objective, and a rationale for measuring target-operator provenance.

8.5 Design implications for subsequent restoration studies

Future models can jointly control image fidelity, measurement-consistent structure, and evaluation under domain shift, while separating training targets from independently validated physical quantities. Candidate objectives include fringe-band spectral preservation, reliable orientation maps, spatial-contrast retention, and repeat-consensus descriptors. These require a confidence mask that distinguishes stable acquisition structure from stochastic speckle. A direct comparison of restoration-only and dual-reference training is an appropriate next experiment, with matched backbones and locked board splits. The present manuscript intentionally supplies no performance predictions for these untrained alternatives.

9 Limitations and reproducibility boundaries

The dataset has six physical specimens, strong class imbalance, and a five-class endpoint predictable from excitation frequency. Sample-level repeats provide many image observations, but they do not create thousands of independent physical boards. The historical averaged-proxy operator has incomplete provenance, and some earlier image-only classifier transfers did not independently establish unseen-board denoiser training. The

corrected five-seed stress result measures one in-distribution setting; the matched U-Net experiment uses one denoiser seed; NAFNet illustrates a specific early failure. These protocols are explicitly separated to preserve valid interpretation.

The nested endpoint reuses verified historical components and provides public aggregate tables, board/seed contrasts, and artifact hashes. The public validator can check their internal consistency and recorded leakage outcomes, but cannot independently rerun private-corpus preprocessing, model inference, or training without source images, checkpoints, and sample-level logits. The six-board bootstrap/sign-flip outputs are descriptive. Future external replication across more specimens, acquisition sessions, and an independently established physical observable would strengthen causal and population-level conclusions.

10 Conclusion

The effective contribution of channel attention is determined by an interaction among gate variability, normalization geometry, downstream weights, residual paths, and supervision. Exact normalization invariances exist under restricted scaling conditions; outside those conditions, learned compensation and small end-to-end sensitivity provide more general explanations for apparently invisible gates. The corrected experiments further show that the optimal denoising configuration depends on the evaluation question. Most decisively, a nested frequency-conditional comparison finds that Raw images provide greater conditional predictive utility than either evaluated DnCNN-ECA view across the locked six-board study, while the absolute incremental-value criterion remains inconclusive. A scientifically informative ESPI restoration evaluation must therefore report its reconstruction target, its non-image baseline, its physical hold-out unit, and the downstream quantity actually measured.

Reproducibility and manuscript status

This is an extended review manuscript assembled from completed, audited aggregate outcomes. The figures displaying numerical results are redrawings of reported public tables. Figure 1 reproduces examples from the research team's earlier manuscript. Raw images, trained checkpoints, and sample-level prediction tensors are not distributed with this manuscript. Proposed physics-aware restoration objectives are future work. For eventual blinded conference submission, source-code links, acknowledgments, and project metadata must be handled under the target venue's anonymity rules.

A Extended mathematical derivations and boundary cases

A.1 Batch-dependent ECA gates break simple BN scale invariance

Let a BN reduction set comprise n scalar activations h_i across different images/spatial locations, with positive gates $a_i = a(h_i, \ldots)$. Define $z_i = a_i h_i$. Then the normalized output uses

$$\mu_z = \frac{1}{n}\sum_i a_i h_i, \qquad v_z = \frac{1}{n}\sum_i (a_i h_i - \mu_z)^2. \tag{30}$$

If $a_i = a$ is shared, $\mu_z = a\mu_h$ and $v_z = a^2 v_h$, giving the restricted invariance in Sec. 5. If the gates vary with the example, $\mu_z \neq a\mu_h$ for any single common a in general. For a simple two-value example, $h = (-1, 1)$, $a = (1/2, 1)$: the gated vector is $(-1/2, 1)$, whose mean is $1/4$, not zero. The dependence of the gate on h adds further covariance terms to v_z. At inference, fixed BN running statistics eliminate even the possibility of instantaneous recomputation-based cancellation. Thus exact BN invariance, static-gate kernel reparameterization, and learned functional equivalence are three distinct mechanisms.

A.2 Group-wise scaling and a counterexample

Take one GN group with two channels and one spatial element each, $h = (1, 3)$. Up to ϵ, the normalized vector is $(-1, +1)$. If gates are $(1, 1/3)$, the gated vector becomes $(1, 1)$ and its variance becomes zero. Any $\epsilon > 0$ gives a normalized vector $(0, 0)$, which is manifestly different. Hence GN does not remove arbitrary channel-wise gates; its invariance is to common positive scaling over the entire group. With GroupNorm(8) and 32 channels, ECA's per-channel gates may change within-group ratios even if each gate is nearly constant across data samples.

A.3 Bounded perturbations for gates near group means

Let $g_c(X) = \overline{g}_{\mathcal{G}}(X)(1 + \delta_c(X))$ within one group, $|\delta_c(X)| \leq \rho < 1$. After removing the common scale $\overline{g}_{\mathcal{G}}$, the remaining activations are $(1 + \delta_c)h_c$. GN is a smooth function away from zero group variance; its local sensitivity is bounded by a constant proportional to $(v_{\mathcal{G}} + \epsilon)^{-1/2}$ and additional variance-dependent terms. Therefore a small ρ can yield a small difference if group variance is bounded away from zero. Near a degenerate group, ϵ controls the denominator but the sensitivity can still be large. This explains why a blanket "GN cancels ECA" statement is unsafe and why grouping, gate heterogeneity, and the feature variance distribution should be measured together.

A.4 Approximate downstream nullspace

Fix an input X and linearize a downstream map Q around $Z_0 = W\overline{G}H$:

$$Q(Z_0 + \delta Z) = Q(Z_0) + J_Q(Z_0)\delta Z + O(\|\delta Z\|^2), \tag{31}$$

with $\delta Z = W\Delta G(X)H(X)$. If $J_Q(Z_0)W\Delta G(X)H(X) \approx 0$, gate-induced changes fall into a local low-sensitivity direction. This requires no literal inverse scaling of γ or β, and can occur in both BN and GN networks. A falsifying experiment measures the alignment of the gate perturbation with the singular vectors of the input-to-output Jacobian or estimates finite differences under standardized gate interventions.

A.5 Residual cancellation and false assurance from absolute differences

The normalization $\mathcal{D}_1$ in Eq. (8) can be small for two distinct reasons. In one, the residual predictions are genuinely equivalent across evaluated inputs. In the other, both residual predictions are small relative to X but differ proportionally in relevant spatial regions. A supplementary diagnostic is the relative residual deviation

$$\mathcal{D}_{\text{rel}} = \frac{\|R_1(X) - R_0(X)\|_2}{\|R_0(X)\|_2 + \epsilon_R}, \tag{32}$$

reported alongside spatially localized, class-conditioned differences. A low average pixel distance is insufficient to rule out changes in rare-mode morphology, particularly in a severely imbalanced five-class task.

A.6 Observational and intervention equivalence

Two networks may have $f_1(X) \approx f_0(X)$ on the audited inputs even when $\theta_1 \neq \theta_0$. A gate ablation performed *after training* evaluates an intervention on a fixed fitted model, and its result need not match the behavior of a separately retrained no-ECA model. To identify compensation, one can intervene separately on the ECA gate and on the next convolution/normalization. If the no-gate intervention changes output while joint removal and reparameterization restore it, the experiment supports a reparameterization explanation. A complete mechanism attribution would additionally require training-trajectory or parameter mapping evidence.

B Detailed statistical reporting ledger

B.1 Six-board primary increments

B.2 Class support and balance

The fixed five-class Macro-F1 assigns equal weight to all classes within each board, and the primary board-balanced summary assigns equal weight to each physical board. This reporting choice is more informative than global accuracy under the severe support imbalance. The condition-balanced sensitivity additionally checks whether boards with numerous repeated acquisitions of a few excitation conditions drive the direction of the paired contrast.

B.3 Confidence-interval interpretation

The interval $[0.03141, 0.06840]$ for Raw–V4R is formed by resampling six observed board-level contrasts (10,000 replicates, seed 42); $[0.03514, 0.10917]$ is the corresponding Raw–V5R interval. These intervals are compact descriptions of variation among the observed specimens. Because the boards form a small and correlated physical sample rather than a random draw from a specified board superpopulation, the coverage properties needed for universal population inference are not established. Exact sign flips enumerate 64 sign assignments and likewise remain descriptive in the absence of randomized assignment/exchangeability conditions.

C Prospective protocol for physically interpretable restoration

This section states a proposed follow-up that has *not* been validated by completed training. It is included to make the negative and conditional findings experimentally actionable rather than to imply an existing solution.

C.1 Repeat-consensus reference with disjoint roles

For each acquisition condition with adequate technically valid real repeats, choose disjoint roles: one input repeat, a set of reconstruction-reference repeats, and an independent set of structural-reference repeats. A leave-one-out or disjoint repeat-consensus statistic can characterize fringe-band energy and orientation stability. Technical eligibility, registration choice, and specimen-level split policy should be frozen before learning. The archived Protocol V2 proposed one input repeat, four reconstruction references, and four structural references for conditions with at least nine valid acquisitions; this was an interface specification, not a completed success claim.

C.2 A candidate multiobjective design

Let $\widetilde{X}$ denote a reconstruction proxy and S_C a structurally reliable statistic estimated from disjoint repeats. A candidate loss is

$$\begin{aligned}\mathcal{L}(D; X, \widetilde{X}, S_C) = {} & \lambda_r \mathcal{L}_{\text{rec}}(D(X), \widetilde{X}) && (33)\\ & + \lambda_f \mathcal{L}_{\text{fringe}}(D(X), S_C) && (34)\\ & + \lambda_o \mathcal{L}_{\text{orientation}}(D(X), S_C) && (35)\\ & + \lambda_i \mathcal{L}_{\text{identity}}(D(\widetilde{X}), \widetilde{X}). && (36)\end{aligned}$$

In contrast to a naive single-shot spectral loss, S_C should encode repeat-consistent structure rather than all of the observed noise. The confidence mask must be computed exclusively from development data. The new method would require matched reconstruction-only controls and a complete checkpoint-selection protocol that never consults held-out-board labels.

C.3 Required evaluation gates

A valid future study should report (i) reconstruction and output-range sanity tests, (ii) nonconstant output and gradient/fringe retention, (iii) identity and randomized-input controls, (iv) a validation-only checkpoint rule, (v) true held-out-board denoiser *and* classifier evaluation, (vi) frequency-only and nearest-frequency comparators, and (vii) paired board-first task differences. The previously attempted set-valued SET-C and SET-D implementations stopped at early output-coherence/safety events before completing their frozen budgets; these events do not establish method infeasibility. They should remain part of the engineering audit history rather than a performance table.

Table 6: Board-level frequency-conditional Macro-F1 increments (seed-averaged within board). These are direct values from the frozen paired-audit table.

Board	Raw increment	V4R increment	V5R increment	Raw–V4R	Raw–V5R
C01	+.047259	−.023479	+.000695	+.070738	+.046564
C02	−.011488	−.069391	−.003221	+.057903	−.008267
C03	−.043435	−.067855	−.135509	+.024420	+.092074
W01	+.077612	+.039852	−.031632	+.037760	+.109244
W02	+.112323	+.026702	−.022049	+.085621	+.134372
W03	+.019924	−.002054	−.059924	+.021978	+.079849
Mean	+.033699	−.016037	−.041940	+.049737	+.075639

Table 7: Numbers of real repeats per class in the frozen six-board endpoint, aggregated from published class-support rows.

Class	Number of repeats
`1_1H`	755
`1_1T`	875
`1_2`	233
`2_1`	423
`higher`	9420
Total	11706

D Additional proof sketch: what a frequency-conditioned score does and does not measure

For a strictly proper probabilistic loss (such as logarithmic score), the Bayes-optimal expected risk difference between $p(Y \mid F)$ and $p(Y \mid F, V)$ can be related to conditional information under the assumed data-generating distribution. Macro-F1 is neither additive over examples nor a strictly proper scoring rule. The operational difference Δ_j of Eq. (25) is therefore not a mutual-information estimator. It depends on class weighting, thresholding/argmax conventions, chosen learner, and available training data. A negative Δ_j may result from estimation variance, fusion-model limitations, domain shift, or genuinely harmful feature transformations. Because Raw and denoised views share labels and images, paired comparisons reduce some sources of variance but do not identify a physical mechanism of representation loss. This is why the physical interpretation is stated as an independent future validation question.

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